TP05 – Apache Kafka

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Q1: What problem does Apache Kafka solve in a data-streaming architecture?
A1:

- It connects many data producers to many consumers without direct links between them.
- It handles continuous real-time streams of events at high volume.
- It decouples senders and readers: they do not need to run at the same time or speed.

Q2: Define producer, consumer, broker, topic, partition, offset, and consumer group.
A2:

- **Producer**: an application that writes events to a topic.
- **Consumer**: an application that reads events from a topic.
- **Broker (Node)**: a Kafka server that stores data and serves producers and consumers.
- **Topic**: a named stream of events (e.g., `campus-sensors`).
- **Partition**: one ordered log that is a part of a topic.
- **Offset**: the position number of an event inside a partition.
- **Consumer group**: consumers sharing the same `group_id`, who split the topic's partitions between them.

Q3: What is the difference between a Kafka topic and a partition?
A3:

- A topic is the logical name of the stream.
- A partition is a physical, ordered part of that topic.
- A topic has one or more partitions.
- Order is guaranteed only inside a partition.
- Partitions allow several consumers to read in parallel.

Q4: Why does the consumer use `auto_offset_reset='earliest'`? When does this setting apply?
A4:

- It makes the consumer start from the oldest available message.
- It applies only when the group has no committed offset (a new group), or when the committed offset no longer exists.
- If the group already has a committed offset, the consumer continues from it and the setting is ignored.

Q5: What is the purpose of the consumer group ID? What happens if a new group ID is used?
A5:

- It identifies the consumer group, so Kafka can share partitions among its members.
- Kafka stores one committed offset per partition for each group.

- With a new group ID, there are no saved offsets, so `auto_offset_reset` applies (with `earliest`, the consumer reads from the beginning).
- The new group reads independently of the other groups.

Q6: Why is JSON a suitable format for this exercise? Give one limitation.
A6:

- It is human-readable and easy to debug.
- It is simple to produce and parse in Python (`json` module).
- It needs no schema setup.
- **Limitation**: it has no enforced schema or types, and it is larger and slower than binary formats (Avro, Protobuf). A missing field is only detected when reading.

Q7: Explain how Docker Compose simplifies deployment compared with starting services manually.
A7:

- One YAML file describes all services, ports, and settings.
- One command (`docker compose up -d`) starts everything, and `down` stops it.
- The setup is reproducible on any machine without manual installation.
- Manually, several long `docker run` commands and the network configuration must be written and repeated each time.

Q8: What are the limitations of this single-broker educational setup compared with a production Kafka cluster?
A8:

- if the broker stops, everything stops.
- Replication factor is limited to 1, so there are no data copies.
- Limited by the CPU, disk, and network of one machine.
- No security configuration (authentication, encryption) and no monitoring.